

```
=====
=====
FILE: src/App.tsx
=====
=====
```

```
import './index.css';
```

```
import type { ReactElement } from 'react';
import { Routes, Route, Navigate } from 'react-router-dom';
```

```
import { ThemeProvider } from './context/ThemeContext';
```

```
import { AuthProvider } from './context/AuthContext';
```

```
import { useAuth } from './hooks/useAuth';
```

```
import Navbar from './Components/Navbar';
```

```
import Home from './pages/Home';
```

```
import Dashboard from './pages/Dashboard';
```

```
import Trade from './pages/Trade';
```

```
import Stake from './pages/Stake';
```

```
import WalletDashboard from './pages/WalletDashboard';
```

```
import Leaderboard from './pages/Leaderboard';
```

```
import Account from './pages/Account';
```

```
import Onboarding from './pages/Onboarding';
```

```
const ProtectedRoute = ({ children }: { children
```

```

: ReactElement }) => {
  const { user, loading } = useAuth();
  if (loading) {
    return <div className="px-8 py-6">Checking s
ession...</div>;
  }
  if (!user) {
    return <Navigate to="/Account" replace />;
  }
  return children;
};

```

```

function App() {
  return (
    <ThemeProvider>
      <AuthProvider>
        <>
          <Navbar />
          <main className="mt-10">
            <Routes>
              <Route path="/" element={<Home />}
/>
              <Route
                path="/Dashboard"
                element={
                  <ProtectedRoute>
                    <Dashboard />
                  </ProtectedRoute>
                }
              />
              <Route

```

```
        path="/Trade"
        element={
            <ProtectedRoute>
                <Trade />
            </ProtectedRoute>
        }
    />
</Route>
<Route
    path="/Stake"
    element={
        <ProtectedRoute>
            <Stake />
        </ProtectedRoute>
    }
/>
</Route>
<Route
    path="/Wallet-Dashboard"
    element={
        <ProtectedRoute>
            <WalletDashboard />
        </ProtectedRoute>
    }
/>
</Route>
<Route
    path="/Leaderboard"
    element={
        <ProtectedRoute>
            <Leaderboard />
        </ProtectedRoute>
    }
/>
```

```

        <Route
          path = "/Onboarding"
          element = {
            <ProtectedRoute>
              <Onboarding/>
            </ProtectedRoute>
          }
        />
        <Route path="/Account" element={<Account />} />
        <Route path="*" element={<Navigate to="/" replace />}} />
      </Routes>
    </main>
  </>
</AuthProvider>
</ThemeProvider>
);
}

export default App;

```

```

=====
=====
FILE: src/Components/Button.tsx
=====
=====

```

```
import type buttonProps from '../types/Button';
//Retrieve button props from the buttonProps interface
```

```
const Button: React.FC<buttonProps> = ({children, variant, className, size, onClick, type, disabled}) => {
    //Creates a forward component so that the children can be passed like a regular button: <Button> text</Button>.
```

```
    //Receives button props - text, variant, className, size
    return (
        <button onClick = {onClick} className={`btn btn-${variant} btn-${size} ${className}`}
        type = {type} disabled={disabled}>
            {children}
        </button>
    );
};
```

```
export default Button;
```

```
=====
=====
```

```
FILE: src/Components/CryptoChart.tsx
```

```
=====
=====

import { useState } from "react";
import SearchBar from "../SearchBar";
import { TradingViewChart } from "../TradingChart";
import { coinNameToTicker } from "../data/tickers";
import { useFetchCoinMetrics } from "../utils/FetchCoinMetrics";

const CryptoChart = () => {
  const [coin, setCoin] = useState("bitcoin");

  const handleSearch = (value: string) => {
    setCoin(value.toLowerCase());
  };

  const logoToken = import.meta.env.VITE_LOGO_API;
  const ticker = coinNameToTicker[coin];

  const { data, loading, error } = useFetchCoinMetrics(coin);

  // --- ERROR STATE: Unknown Coin Ticker ---
  if (!ticker) {
    return (
      <div className="p-8 text-center bg-surface min-h-screen text-primary">
```

```

        <SearchBar Search={handleSearch} />
        <p className="mt-8 text-xl text-danger">
          ! Unknown cryptocurrency: <strong>{coin}</strong>
        </p>
        <p className="text-muted">Try a common coin like "bitcoin" or "ethereum".</p>
      </div>
    );
  }

```

```

// --- LOADING/ERROR STATES ---
if (loading) {
  return (
    <div className="p-8 text-center bg-surface min-h-screen text-primary">
      <SearchBar Search={handleSearch} />
      <div className="mt-20 text-xl animate-pulse">Loading data for {coin}...</div>
    </div>
  );
}

```

```

if (error || !data) {
  return (
    <div className="p-8 text-center bg-surface min-h-screen text-primary">
      <SearchBar Search={handleSearch} />
      <div className="mt-20 text-xl text-danger">

```

✗ Fetch failed: {error || 'No data returned'}

```
urned from API.'}
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

```
// --- RENDER DASHBOARD ---
```

```
// Determine color for 24h change (Success/Danger is common in trading UI)
```

```
const changeColor = data.change24h >= 0 ? 'text-success' : 'text-danger';
```

```
const changeSign = data.change24h >= 0 ? '↑' : '↓';
```

```
return (
```

```
<div className="min-h-screen bg-surface text-primary p-4 sm:p-8 font-sans">
```

```
<div className="max-w-7xl mx-auto">
```

```
  {/* 1. Search Bar (Top Control) - Assuming SearchBar has minimal styling */}
```

```
<div className="mb-8">
```

```
<SearchBar Search={handleSearch} />
```

```
</div>
```

```
  {/* 2. Header and Price Metrics */}
```

```
<div className="flex items-center mb-6 space-x-4 border-b border-divider pb-4">
```

```
<img
```

```
  src={`https://img.logokit.com/crypto
```



```

/${ticker}?token=${logoToken}`}
    alt={`${coin} logo`}
    className="w-10 h-10 rounded-full shadow-md"
  />
  <div>
    <h1 className="text-2xl font-semibold capitalize">{coin}</h1>
    <p className="text-sm text-muted">{ticker} / USD</p>
  </div>

  {/* Current Price and 24h Change */}
  <div className="ml-auto flex items-end space-x-4">
    <p className="text-4xl font-bold">${data.price.toLocaleString('en-US', { minimumFractionDigits: 2, maximumFractionDigits: 8 })}</p>
    <p className={`text-lg font-medium ${changeColor} mb-0.5`} >
      {changeSign} {data.change24h.toFixed(2)}%
    </p>
  </div>

  {/* 3. Main Chart (Placeholder) */}
  <div className="bg-card rounded-lg p-2 shadow-lg h-[450px] mb-8 border border-divider">
    <TradingViewChart coin={coin} mode="candlestick" />
  </div>

```

```
</div>
```

```
{/* 4. Key Metrics Grid */}
```

```
<div className="grid grid-cols-2 md:grid  
-cols-4 gap-4">
```

```
  {/* Market Cap */}
```

```
  <MetricCard
```

```
    title="Market Cap"
```

```
    value={`$$${(data.marketCap / 1e9).to  
Fixed(2)}B`}
```

```
    tooltip={`$$${data.marketCap.toLocale  
String()}`}
```

```
  />
```

```
  {/* 24h Volume */}
```

```
  <MetricCard
```

```
    title="24h Volume"
```

```
    value={`$$${(data.volume24h / 1e9).to  
Fixed(2)}B`}
```

```
    tooltip={`$$${data.volume24h.toLocale  
String()}`}
```

```
  />
```

```
  {/* Fully Diluted Valuation (FDV) */}
```

```
  <MetricCard
```

```
    title="FDV"
```

```
    value={data.fullyDilutedValuation  
      ? `$$${(data.fullyDilutedValuation  
/ 1e9).toFixed(2)}B`
```

```
      : 'N/A'
```

```
    }
```

```

        tooltip={data.fullyDilutedValuation
        ? `$$${data.fullyDilutedValuation.to
oLocaleString()}`
        : 'Not available'
    }
  />

```

```

    {/* Circulating Supply */}
    <MetricCard
      title="Circulating Supply"
      value={`$${(data.circulatingSupply /
1e6).toFixed(2)}M`}
      tooltip={data.circulatingSupply.toLo
caleString()}
    />
  </div>

```

```

    {/* 5. Additional Supply Data (Muted Foo
ter) */}
    <div className="mt-8 text-xs text-muted
text-right">
      Total Supply: {data.totalSupply?.toL
ocaleString() || 'Unlimited/N/A'}
    </div>

```

```

    </div>
  </div>
);
};

```

```

// Helper component for clean metric display

```

```

const MetricCard = ({ title, value, tooltip }: {
  title: string, value: string, tooltip: string }
) => (
  <div className="bg-card p-4 rounded-md shadow-
sm border border-divider hover:shadow-lg transit
ion duration-150 relative group">
    <p className="text-sm font-medium text-muted
">{title}</p>
    <p className="text-xl font-semibold mt-1">{v
alue}</p>
    { /* Tooltip implementation remains the same
for functionality */ }
    <span className="absolute bottom-full left-1
/2 transform -translate-x-1/2 mb-2 px-3 py-1 bg-
tooltip text-xs text-tooltip-text rounded opacit
y-0 group-hover:opacity-100 transition-opacity d
uration-300 pointer-events-none whitespace-nowra
p">
      {tooltip}
    </span>
  </div>
);

export default CryptoChart;

```

```

=====
=====
FILE: src/Components/DMLMToggle.tsx
=====

```

```
=====

import { useTheme } from "../hooks/useTheme";
import Button from "../Button";
export const Toggle = () => {
  const {isDark, toggleTheme} = useTheme();
  return (
    <Button
      onClick={toggleTheme}
      size="lg"
      type="button"
      variant="secondary"
    >
      {isDark ? <img className = "w-5 h-5" src =
"/public/sun.svg"/> : <img className = "w-5 h-5
" src = "/public/moon.svg"/>}
    </Button>
  );
};
```

```
=====
=====
FILE: src/Components/Navbar.tsx
=====
=====
```

```
import Button from "../Button";
import { useNavigate, useLocation } from "react-
router-dom";
```

```
import { Links } from "../lib/links";
import { Toggle } from "../DMLMToggle";
```

```
const Navbar = () => {
  const navigate = useNavigate();
  const location = useLocation();
  const {pathname} = location;
  return (
```

```
<nav className="bg-neutral-primary w-full z-20
top-0 start-0 ">
  <div className="max-w-7xl flex flex-wrap items
-center justify-between mx-auto p-4">
    <a href="/" className="flex items-center spa
ce-x-3 rtl:space-x-reverse">
```

```
      <span className="self-center text-xl tex
t-heading font-semibold whitespace-nowrap">Block
Finance</span>
    </a>
```

```
    <button data-collapse-toggle="navbar-default
" type="button" className="inline-flex items-cen
ter p-2 w-10 h-10 justify-center text-sm text-bo
dy rounded-base md:hidden hover:bg-neutral-secon
dary-soft hover:text-heading focus:outline-none
focus:ring-2 focus:ring-neutral-tertiary" aria-c
ontrols="navbar-default" aria-expanded="false">
      <span className="sr-only">Open main menu
    </span>
    <svg className="w-6 h-6" aria-hidden="tr
```

```
ue" xmlns="http://www.w3.org/2000/svg" width="24"
" height="24" fill="none" viewBox="0 0 24 24"><p
ath stroke="currentColor" strokeLinecap="round"
strokeWidth="2" d="M5 7h14M5 12h14M5 17h14"/></s
vg>
```

```
</button>
```

```
<div className="hidden w-full md:block md:w-
auto" id="navbar-default">
```

```
<ul className="font-medium flex flex-col p-4
md:p-0 mt-4 border rounded-lg md:flex-row md:sp
ace-x-8 md:mt-0 md:border-0">
```

```
{Links.map((item) => (
```

```
<li key={item.link}>
```

```
<Button
```

```
onClick={() => navigate(item.link)}
```

```
type="button"
```

```
variant="secondary"
```

```
size="md"
```

```
//Created a template string that checked
whether the pathname is equal to the link and c
hanged the color of the text accordingly.
```

```
className={` ${pathname === item.link ?
'text-gray-800' : ''} `}
```

```
>
```

```
{item.name}
```

```
</Button>
```

```
</li>
```

```
)))
```

```
<Toggle/>
```

```
</ul>
    </div>
  </div>
</nav>
```

```
    );
  };
export default Navbar
```

```
=====
=====
FILE: src/Components/NewsCard.tsx
=====
=====
```

```
interface NewsCardProps {
  title: string;
  description: string | null;
  link: string;
  imageUrl: string | null;
  pubDate: string;
  source: string;
}
```

```
const NewsCard = ({ title, description, link, imageUrl, pubDate, source }: NewsCardProps) => {
```



```

const formatDate = (dateString: string) => {
  const date = new Date(dateString);
  const now = new Date();
  //Difference in hours
  const diffInHours = Math.floor((now.getTime(
) - date.getTime()) / (1000 * 60 * 60));
  //Use if statements to determine how recent
a post is for the user - improving user experien
ce
  if (diffInHours < 1) return 'Just now';
  if (diffInHours < 24) return `${diffInHours}
h ago`;
  const diffInDays = Math.floor(diffInHours /
24);
  if (diffInDays < 7) return `${diffInDays}d a
go`;
  return date.toLocaleDateString('en-US', { mo
nth: 'short', day: 'numeric' });
};

```

```

return (
  <a
    href={link}
    target="_blank"
    rel="noopener noreferrer"
    className="
      block rounded-xl border border-[var(--bord
er-color)]
      bg-[var(--surface-color)] text-[var(--text
-color)]
      no-underline overflow-hidden hover:shadow-

```

md

```
    transition-shadow duration-200 group
```

```
>
```

```
    <div className="w-full h-40 overflow-hidd  
en ">
```

```
    <img
```

```
    //If image url isn't present then use the  
    backup image
```

```
        src={imageUrl || "/backup.webp"}
```

```
        alt={title}
```

```
        //Typical tailwindcss animation for ima  
ge upscale on hover
```

```
        className="w-full h-full object-cover g  
roup-hover:scale-105 transition-transform durati  
on-200"
```

```
    />
```

```
</div>
```

```
    <div className="p-4">
```

```
        <div className="flex items-center justif  
y-between mb-2">
```

```
            <span className="text-xs font-medium u  
ppercase text-[var(--muted-text-color)]">  
                {source}
```

```
            </span>
```

```
            <span className="text-xs text-[var(--m  
uted-text-color)]">  
                {formatDate(pubDate)}
```

```

        </span>
      </div>
      <h3 className="text-sm font-semibold mb-
2 line-clamp-2">
        {title}
      </h3>
      { /*If description then showcase it */}
      {description && (
        <p className="text-xs text-[var(--mute
d-text-color)] line-clamp-2">
          {description}
        </p>
      )}
    </div>
  </a>
);
};

```

```

export default NewsCard;

```

```

=====
=====
FILE: src/Components/SearchBar.tsx
=====
=====

```

```

import { useState } from 'react';
import Button from './Button';

```

```
interface SearchBarProps {  
  Search: (coin: string) => void;  
}
```

```
const SearchBar: React.FC<SearchBarProps> = ({ S  
earch }) => {  
  const [query, setQuery] = useState("");  
  
  const handleClick = () => {  
    Search(query.trim());  
  };  
};
```

```
  return (  
    <>  
      <form  
        className="max-w-md mx-auto"  
        onSubmit={(e) => {  
          e.preventDefault(); // prevents page rel  
oad  
          handleClick();  
        }}  
      >  
        <label htmlFor="search" className="block m  
b-2.5 text-sm font-medium text-heading sr-only">  
          Search  
        </label>  
  
        <div className="relative">  
          <div className="absolute inset-y-0 start  
-0 flex items-center ps-3 pointer-events-none">
```

```

<svg
  className="w-4 h-4 text-body"
  aria-hidden="true"
  xmlns="http://www.w3.org/2000/svg"
  width="24"
  height="24"
  fill="none"
  viewBox="0 0 24 24"
>
  <path stroke="currentColor" strokeLine
necap="round" strokeWidth="2" d="m21 21-3.5-3.5M
17 10a7 7 0 1 1-14 0 7 7 0 0 1 14 0Z" />
</svg>
</div>

```

```

<input
  type="search"
  value={query}
  onChange={(e) => setQuery(e.target.val
ue)}
  className="block w-full p-3 ps-9 bg-ne
utral-secondary-medium text-heading text-sm rou
nded-base shadow-xs placeholder:text-body"
  placeholder="Search"
/>

```

```

<Button
  variant = "secondary"
  size = "sm"
  type="submit"
  className="absolute end-0 bottom-0

```

```
border border-transparent focus:ring-4 focus:ring-brand-medium shadow-xs font-medium leading-5 rounded text-xs px-3 py-1.5"
```

```
    >  
      Search  
    </Button>  
  </div>  
</form></>
```

```
);  
};
```

```
export default SearchBar;
```

```
=====
```

```
=====
```

```
FILE: src/Components/Tooltip.tsx
```

```
=====
```

```
=====
```

```
import { HelpCircle } from 'lucide-react';  
import { useState, type ReactNode } from 'react';  
;
```

```
interface TooltipProps {  
  text: string;  
  children?: ReactNode;  
  position?: 'top' | 'bottom';  
}
```

```

const Tooltip = ({ text, children, position = 'top' }: TooltipProps) => {
  const [open, setOpen] = useState(false);
  const tooltipPositionClass = position === 'top'
    ? 'bottom-full mb-2' : 'top-full mt-2';

  const accessibleLabel =
    typeof children === 'string' && children.trim() !== '' ? `More info about ${children}` : 'More info';

  return (
    <span className="relative inline-flex align-baseline">
      <span
        role="button"
        tabIndex={0}
        className="inline-flex items-baseline cursor-help border-b border-dashed border-current bg-transparent p-0"
        onMouseEnter={() => setOpen(true)}
        onMouseLeave={() => setOpen(false)}
        onFocus={() => setOpen(true)}
        onBlur={() => setOpen(false)}
        onClick={() => setOpen(value) => !value}
      >
        {children}
      </span>
      {accessibleLabel}
    </span>
  );
};

```

```

        setOpen((value) => !value);
    }
}
style={{ lineHeight: 'inherit' }}
aria-label={accessibleLabel}
aria-expanded={open}
>
<span className="inline">{children}</span>
n>
    <HelpCircle className="ml-1 inline-block
h-3 w-3" style={{ verticalAlign: 'baseline', po
sition: 'relative', top: '0.05em' }} />
</span>

{open && (
    <span
        role="tooltip"
        className={`pointer-events-none absolu
te left-1/2 z-[70] w-max max-w-xs -translate-x-1
/2 rounded-lg border border-[var(--border-color)]
bg-[var(--surface-color)] px-3 py-2 text-xs sh
adow-xl ${tooltipPositionClass}`}
        style={{ color: 'var(--text-color)' }}
    >
        {text}
    </span>
)}
</span>
);
};

```



```
export default Tooltip;
```

```
=====
=====
FILE: src/Components/TradingChart.tsx
=====
=====
```

```
import {
  createChart,
  ColorType,
  CandlestickSeries,
  LineSeries,
  type CandlestickData,
  type IChartApi,
  type ISeriesApi,
  type UTCTimestamp,
} from 'lightweight-charts';
import { useEffect, useRef, useState } from 'react';
import { fetchCandleData } from '../utils/FetchCandleData';
import { useTheme } from '../hooks/useTheme';

type ChartMode = 'line' | 'candlestick';

interface TradingViewChartProps {
  coin: string;
  mode?: ChartMode;
```

```

onMarketDataChange?: (marketData: {
  lastPrice: number | null;
  change24h: number | null;
}) => void;
}

const TradingViewChart = ({
  coin,
  mode = 'candlestick',
  onMarketDataChange,
}: TradingViewChartProps) => {
  const { isDark } = useTheme();
  const chartContainerRef = useRef<HTMLDivElement
  | null>(null);
  const chartRef = useRef<IChartApi | null>(null
  );
  const candleSeriesRef = useRef<ISeriesApi<'Can
  dlestick'> | null>(null);
  const lineSeriesRef = useRef<ISeriesApi<'Line'
  > | null>(null);
  const [candles, setCandles] = useState<Candles
  tickData<UTCTimestamp>[]>([]);

  useEffect(() => {
    if (!chartContainerRef.current) {
      return;
    }

    const chart = createChart(chartContainerRef.
    current, {
      layout: {

```

```
        background: { type: ColorType.Solid, color: isDark ? '#000000' : '#ffffff' },
        textColor: isDark ? '#e5e7eb' : '#1f1f1f',
    },
    grid: {
        vertLines: { color: isDark ? '#34343a' : '#e5e7eb' },
        horzLines: { color: isDark ? '#34343a' : '#e5e7eb' },
    },
    width: chartContainerRef.current.clientWidth,
    height: 400,
});
```

```
    if (mode === 'candlestick') {
        candleSeriesRef.current = chart.addSeries(CandlestickSeries, {
            upColor: '#26a69a',
            downColor: '#ef5350',
            wickUpColor: '#26a69a',
            wickDownColor: '#ef5350',
            borderVisible: false,
        });
        lineSeriesRef.current = null;
    } else {
        lineSeriesRef.current = chart.addSeries(LineSeries, {
            color: '#2962FF',
            lineWidth: 2,
```

```

    });
    candleSeriesRef.current = null;
  }

  chartRef.current = chart;

  const handleResize = () => {
    if (chartContainerRef.current) {
      chart.applyOptions({ width: chartContainerRef.current.clientWidth });
    }
  };

  window.addEventListener('resize', handleResize);

  return () => {
    window.removeEventListener('resize', handleResize);
    chart.remove();
    chartRef.current = null;
    candleSeriesRef.current = null;
    lineSeriesRef.current = null;
  };
}, [isDark, mode]);

useEffect(() => {
  let mounted = true;

  const loadCandles = async () => {
    setCandles([]);
  };

```

```

    const nextCandles = await fetchCandleData(
coin);
    if (mounted) {
        setCandles(nextCandles);
    }
};

void loadCandles();

return () => {
    mounted = false;
};
}, [coin]);

useEffect(() => {
    if (candles.length === 0) {
        return;
    }

    if (mode === 'candlestick' && candleSeriesRe
f.current) {
        candleSeriesRef.current.setData(candles);
        chartRef.current?.timeScale().fitContent()
;
        return;
    }

    if (mode === 'line' && lineSeriesRef.current
) {
        const lineData = candles.map((candle) => (
{

```

```

        time: candle.time,
        value: candle.close,
    }));
    lineSeriesRef.current.setData(lineData);
    chartRef.current?.timeScale().fitContent()
;
    }, [candles, mode]);

    const latestChartPrice = candles.length > 0 ?
candles[candles.length - 1].close : null;
    const price24hAgo = candles.length > 24 ? cand
les[candles.length - 25].close : null;
    const chartChange24h =
        latestChartPrice !== null &&
        price24hAgo !== null &&
        Number.isFinite(price24hAgo) &&
        price24hAgo > 0
        ? ((latestChartPrice - price24hAgo) / pric
e24hAgo) * 100
        : null;

    useEffect(() => {
        if (!onMarketDataChange) {
            return;
        }

        onMarketDataChange({
            lastPrice: latestChartPrice,
            change24h: chartChange24h,
        });
    });

```

```
    }, [chartChange24h, latestChartPrice, onMarket  
DataChange]);
```

```
    return <div ref={chartContainerRef} style={{ w  
idth: '100%', height: '400px' }} />;  
};
```

```
export type { ChartMode };  
export { TradingViewChart };
```

```
=====
```

```
=====
```

```
FILE: src/context/AuthContext.tsx
```

```
=====
```

```
=====
```

```
import {  
  useEffect,  
  useState,  
  type ReactNode,  
} from 'react';  
import { supabase } from '../lib/supabaseClient'  
;  
import { AuthContext, type AuthContextType } fro  
m './authContextStore';
```

```
export const AuthProvider = ({ children }: { chi  
ldren: ReactNode }) => {  
  const [user, setUser] = useState<AuthContextTy
```

```

pe['user']>(null);
const [loading, setLoading] = useState(true);

useEffect(() => {
  const init = async () => {
    const { data, error } = await supabase.auth
h.getUser();
    if (!error && data.user) {
      setUser(data.user);
    }
    setLoading(false);
  };

  void init();

  const {
    data: { subscription },
  } = supabase.auth.onAuthStateChange((_event,
session) => {
    setUser(session?.user ?? null);
  });

  return () => {
    subscription.unsubscribe();
  };
}, []);

//

```